USL Assignment 1

Implementation of Distance Measures using Python

Name: Aadi Kulshreshth
PRN: 24070126001

Dataset Description

The **Breast Cancer Wisconsin (Diagnostic) dataset** consists of numerical features computed from digitized images of breast mass samples. Each record represents characteristics of cell nuclei present in the image.

- Number of instances: 569
- Number of attributes: 30 numerical features
- Target attribute: Diagnosis (Benign / Malignant)
- Dataset type: Numerical, continuous

For this assignment, the target attribute is excluded and only feature vectors are used for distance computation.

Preprocessing Overview

Before computing distances, the dataset undergoes basic preprocessing:

- Removal of irrelevant columns such as ID and diagnosis
- Standardization of numerical features to avoid scale dominance
- Binary conversion where required for binary distance measures

Theory of Distance Measures

1. Euclidean Distance

Measures straight-line distance between two points.

Formula

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (y_i - x_i)^2}$$

Use case: Continuous numerical data.

2. Manhattan Distance

Measures distance as the sum of absolute differences.

Formula

$$d(x, y) = \sum_{i=1}^{n} |x_i - y_i|$$

Use case: Grid-based paths, robust to outliers.

3. Minkowski Distance

Generalized distance metric.

Formula

$$d(x, y) = \left(\sum_{i=1}^{n} |x_i - y_i|^c \right)^{\frac{1}{c}}$$

- c = 1 → Manhattan
- c = 2 → Euclidean

4. Hamming Distance

Counts mismatches between binary vectors.

Formula

$$d(x, y) = \frac{1}{n} \sum_{n=1}^{n=n} |x_i - y_i|$$

Use case: Binary/categorical data.

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5. Jaccard Distance

Measures dissimilarity between binary sets.

Formula

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Use case: Presence/absence data.